U-Net for Retinal Vessel Segmentation

Project Deep Learning Architectures

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Outline

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Introduction & Motivation

Context:

- Retinal vessel segmentation is crucial for diagnosing diabetic retinopathy and hypertension.
- Manual segmentation is time-consuming and prone to error.

Project Goals:

- **Core Objective:** Implement a robust training pipeline using the **Standard U-Net** on the DRIVE dataset.
- **Secondary Objective:** Extend the model with **Attention Gates** (Attention U-Net) to analyze if adding model complexity improves segmentation performance on fine vascular structures.

The U-Net Architecture

Key Features [1]:

- **Encoder (Left):** Captures context.
- **Decoder (Right):** Enables precise localization.
- **Skip Connections:** Recover spatial information lost during down-sampling.

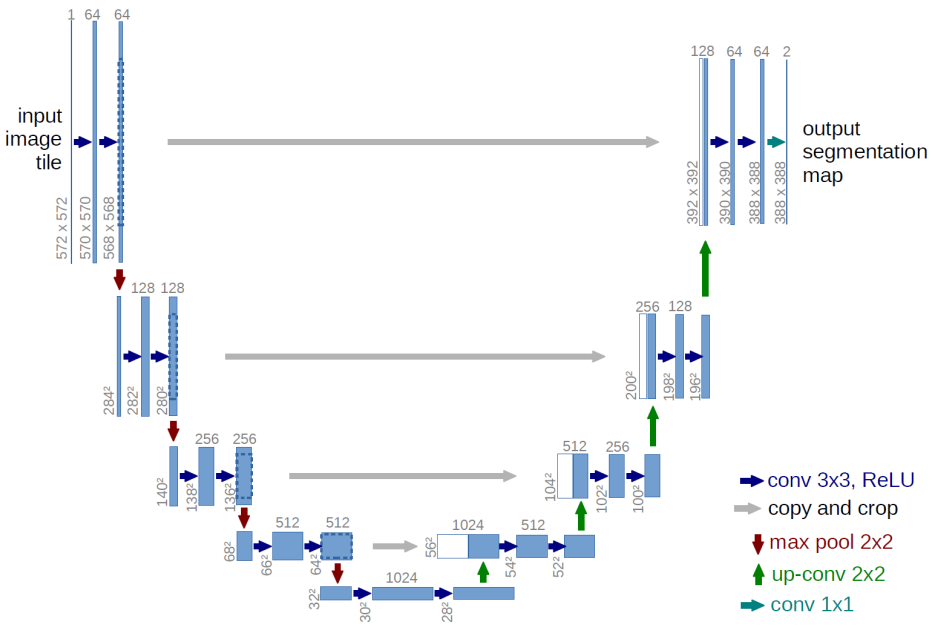

Figure: Standard U-Net Architecture.

Attention U-Net

Concept:

- Integrates **Attention Gates (AGs)** into skip connections.
- Suppresses irrelevant background noise to focus on target structures [2].

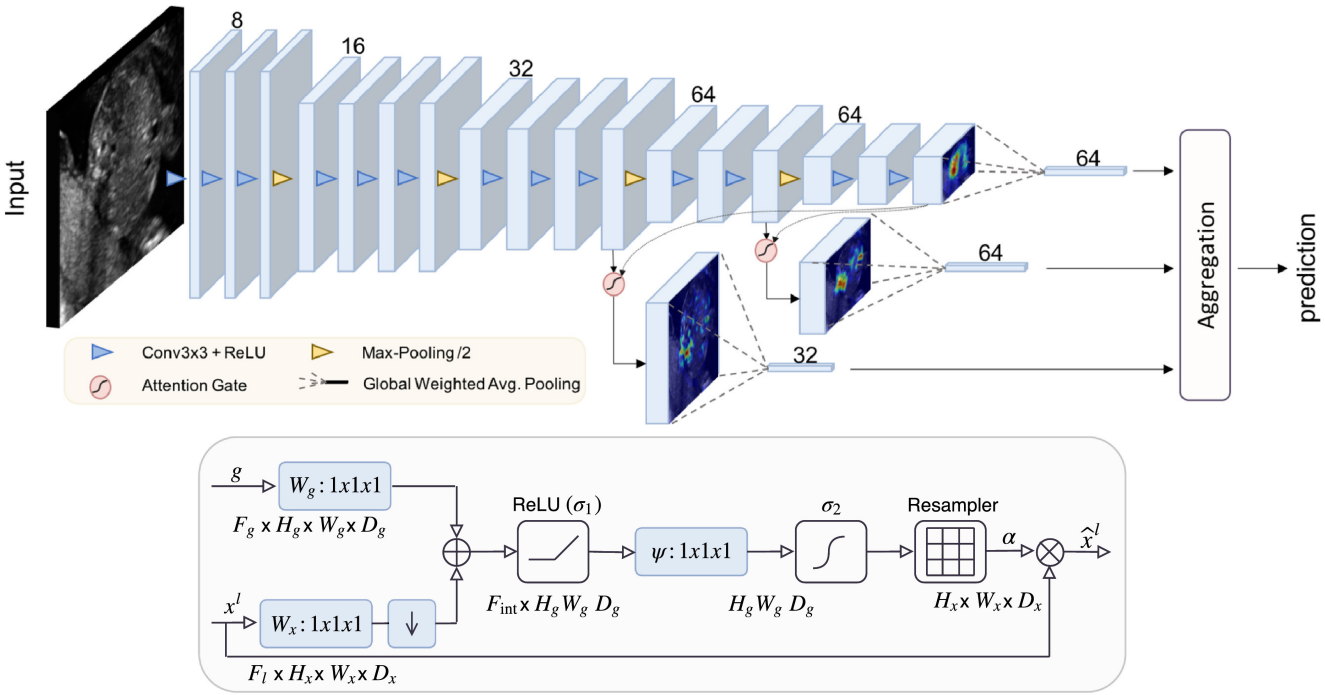

Figure: Top: Attention U-Net. Bottom: Attention Gate mechanism.

Dataset: DRIVE

Digital Retinal Images for Vessel Extraction (DRIVE) [3]:

- **Data:** 40 color fundus images.
- **Split:** 20 Training / 20 Testing.
- **Task:** Binary Segmentation (Vessel vs. Background).

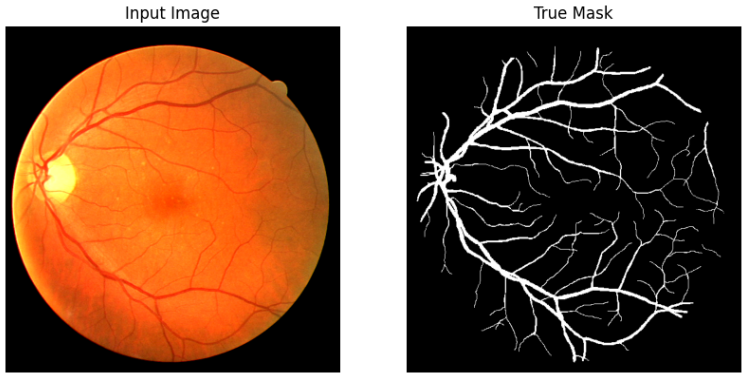

Figure: Input image and Ground Truth Mask.

Implementation Details

Framework: Python & PyTorch

Training Strategy:

- **Data Preprocessing:** Resized to 512×512.
- **Loss Function:** Binary Cross Entropy (BCE) + Dice Loss.
- **Optimizer:** AdamW (Learning rate $1e-4$).
- **Augmentation:** Random rotation, flips, and elastic deformation via Albumentations to prevent overfitting.
- **Hardware & Execution:** Trained for 200 epochs using Apple Silicon (MPS backend) with an Early Stopping patience of 30 epochs.

Quantitative Results

Models were evaluated on the unseen 20-image test set.

Model (200 Epochs)	Accuracy	Sensitivity	Specificity	Dice Score
Standard U-Net	0.9609	**0.8267**	0.9740	**0.7860**
Attention U-Net	0.9585	0.7774	**0.9761**	0.7651

Table: Final Test Set Results. The Standard U-Net achieved higher sensitivity and overall overlap (Dice).

Qualitative Results: Standard U-Net

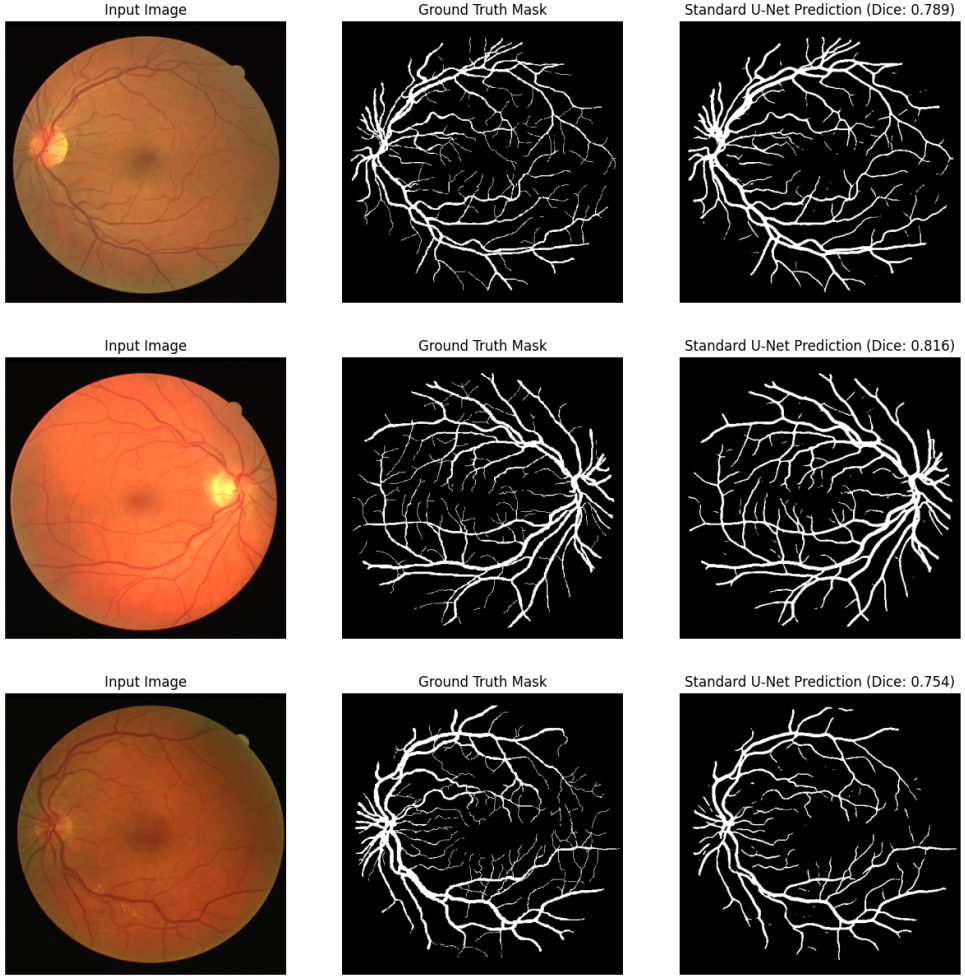

Figure: Input, Ground Truth, and Standard U-Net Prediction.

Qualitative Results: Attention U-Net

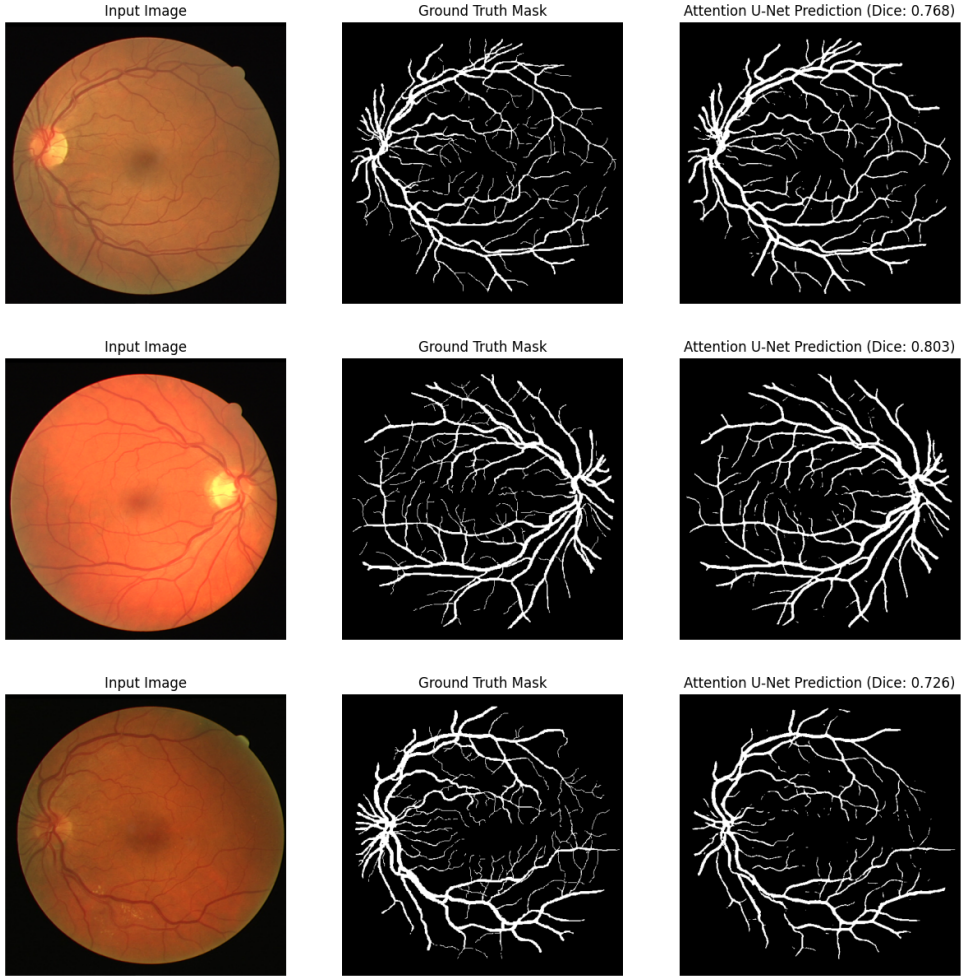

Figure: Input, Ground Truth, and Attention U-Net Prediction. (Note the fragmented predictions on fine vessels).

Comparative Analysis: The Overfitting Phenomenon

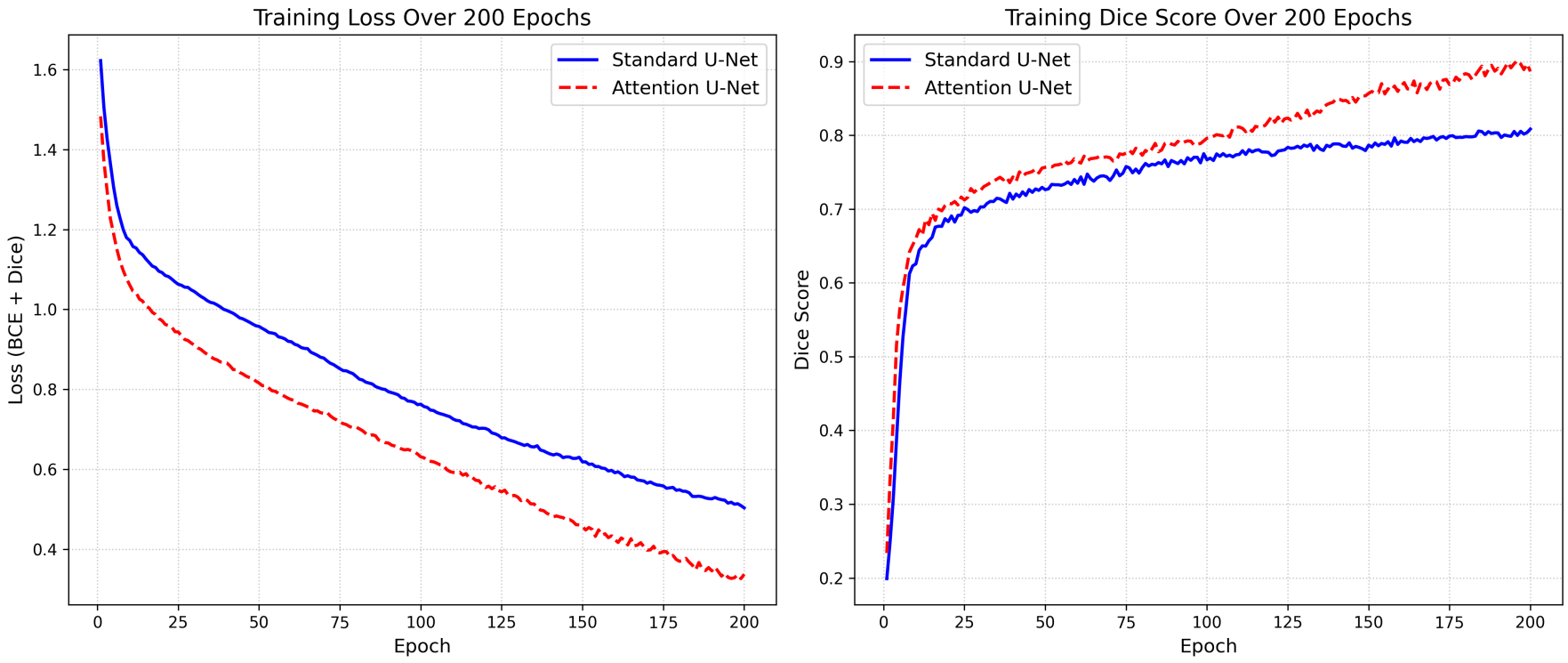

Figure: Learning curves reveal severe overfitting by the Attention U-Net. It memorized the training data ($\sim$ 90% Dice) but failed to generalize to the test set (76.5% Dice).

Conclusion

Summary of Findings:

- Successfully established a robust training pipeline, achieving a strong 0.7860 test Dice score with the Standard U-Net.
- The Standard U-Net effectively generalized to capture fine vascular structures without succumbing to noise.
- Adding architectural complexity (Attention Gates) on a highly constrained dataset (only 20 images) induced severe overfitting.

Key Takeaway:
For tasks with limited data availability, simpler architectures often yield more robust and generalizable medical segmentation models.

Code Repository:
`https://github.com/RanaPadi/deep-learning-biomedical-segmentation.git`

References I

- O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *MICCAI 2015*. Springer, 2015, pp. 234-241.
- O. Oktay et al., "Attention u-net: Learning where to look for the pancreas," *arXiv preprint arXiv:1804.03999*, 2018.
- J. Staal, M. D. Abràmoff, M. Niemeijer, M. A. Viergever, and B. van Ginneken, "Ridge-based vessel segmentation in color images of the retina," *IEEE transactions on medical imaging*, vol. 23, no. 4, pp. 501-509, 2004.